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**S7-200 PLC Programming: Practical Exercise Series (200 Problems)**

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**I. Basic Bit Logic (I, Q, M) - [Problems 1-40]**

1. Switch **I0.0** turns ON Bulb **Q0.0**.
2. Switch **I0.1** turns ON Motor **Q0.1**.
3. **Q0.2** turns ON only if both **I0.0** AND **I0.1** are ON.
4. **Q0.3** turns ON if either **I0.2** OR **I0.3** is ON.
5. Latch **Q0.0** using **I0.0** (Start) and **M0.0** (Memory).
6. Use **I0.1** (NC Stop) to unlatch the above **Q0.0**.
7. Create an Interlock: **Q0.0** cannot be ON if **Q0.1** is ON.
8. Use **I0.0** to **SET** memory bit **M0.0**.
9. Use **I0.1** to **RESET** memory bit **M0.0**.
10. Trigger **Q0.4** using memory bit **M0.0**.
11. Activate Alarm **Q0.5** if **I0.0** and **I0.1** are pressed simultaneously.
12. Use **M0.1** as a Master Enable bit for input **I0.3**.
13. Trigger **Q0.1** using a **Positive Edge** pulse from **I0.4**.
14. Turn OFF **Q0.2** using a **Negative Edge** pulse from **I0.5**.
15. Use one switch **I0.0** to start two motors **Q0.0** and **Q0.1** together.

16. Use **I0.1** to stop both motors simultaneously.
17. Toggle **M0.2** state with a single button **I0.2** (Flip-flop logic).
18. Show "System Ready" on **Q0.3** using memory bit **M0.3**.
19. Use Emergency Stop **I0.6** to reset all active **Q** outputs.
20. Use **I0.7** as a Selector Switch to choose between Auto and Manual modes. *(Problems 21-40: Continue mixing I/O and M bits with complex Logic Gates like NAND, NOR, and XOR)*

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**II. Timers (T) & Timing Logic - [Problems 41-80]**

41. Turn ON **Q0.0** exactly 5 seconds after pressing **I0.0** (Use **T37**).
42. Keep Motor **Q0.1** running for 10 seconds and then stop automatically.
43. Create a Blinking Light on **Q0.2** (1s ON / 1s OFF) using two timers.
44. Press **I0.0** to start **Q0.0** immediately, and **Q0.1** after a 5s delay.
45. Create a 3-second Pulse on **Q0.3** when **I0.2** is triggered.
46. Turn ON **M0.0** using **T37**, then use **M0.0** to trigger **Q0.0**.
47. Use a Retentive Timer (**T1**) to accumulate total machine run time.
48. Trigger Maintenance Alarm **Q0.4** once total run time reaches 100 seconds.
49. Reset all active timers using switch **I0.3**.
50. Create "Running Lights": **Q0.0, Q0.1, Q0.2** turn ON sequentially every 2s. *(Problems 51-80: Vary time constants from ms to minutes and practice cascading timers)*

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**III. Counters (C) & Counting Logic - [Problems 81-110]**

81. Count pulses from **I0.0** using Up-Counter **C1**.
82. Turn ON **Q0.0** when **C1** reaches a count of 10.
83. Reset **C1** using a separate button **I0.1**.
84. Use **I0.2** (Inbound) and **I0.3** (Outbound) for Up/Down counting (**C2**).
85. Activate "Parking Full" light **Q0.1** when **C2** reaches 50.
86. Activate "Parking Empty" light **Q0.2** when **C2** reaches 0.

87. Stop Conveyor **Q0.0** using **M1.0** after counting 12 products.

88. Clear the count value using memory bit **M1.1**.

89. Stop the entire process after a batch count of 5 reaches its target.

90. Make **Q0.5** flash once Counter **C1** exceeds 100.

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**IV. Data Handling & Math (VW, MW) - [Problems 111-160]**

111. Turn ON **Q0.0** if the value in **VW100** is greater than 50.

112. Turn ON **M2.0** if **VW100** and **VW102** are equal.

113. Turn ON **Q0.1** if **VW104** is between 10 and 20.

114. Add 10 to **VW106** and store the result in **VW108** (**ADD_I**).

115. Subtract **VW112** from **VW110** (**SUB_I**).

116. Multiply **VW114** by 2 and display the result.

117. Copy the value from **VW10** to **VW20** using the **MOV** instruction.

118. Reset data by moving 0 to **VW100** when **I0.0** is pressed.

119. Move the current value of **C1** to **VW200** for HMI display.

120. Store the elapsed time of **T37** into **VW210**. *(Problems 121-160: Practice Integer and Floating point math with various memory addresses)*

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**V. Industrial Integration (Mixed Challenges) - [Problems 161-200]**

161. **Auto Conveyor:** Start **Q0.0** with **I0.0**. Stop when **I0.1** (Sensor) counts 10 items.

162. **Fault Indicator:** If Motor trips (**I0.2**), flash Red Light **Q0.1** (1s cycle).

163. **Two-Hand Safety: Q0.0** (Cutter) works only if **I0.0** and **I0.1** are pressed within 0.5s.

164. **Pump Alternator:** Toggle between Pump **Q0.0** and **Q0.1** on every Start pulse.

165. **Staircase Light:** Turn ON **Q0.0** for 30s upon trigger, then auto-OFF.

166. **Bottling Plant:** Stop conveyor **Q0.0** when sensor **I0.1** detects a bottle; fill for 5s.

167. **Sequence Lock: Q0.0** opens only if buttons **I0.0, I0.1, I0.2** are pressed in order.

168. **Service Alarm:** Trigger **Q0.7** after the machine completes 1000 cycles.

169. **Speed Watchdog:** Trigger alarm **Q0.5** if pulses from **I0.0** are < 50 in 10 seconds.

170. **Variable Timer:** Use the value in **VW100** to set the delay for **T37**.

171. **Cooldown Logic:** Fan **Q0.0** stays ON for 10s after the heat sensor **I0.0** turns OFF.

172. **Global Reset:** Emergency **I0.7** resets all Q, T, C, and M bits.

173. **Batching:** Set **M0.0** after 12 items. Trigger **Q1.0** after 10 boxes are filled.

174. **Star-Delta Starter:** Implement Star-Delta transition logic for a motor.

175. **Level Simulation:** Change **VW100** value (10, 20, 30) based on button presses.

176. **Siren Acknowledge:** Siren **Q0.0** sounds on fault; stops only when **I0.1** (ACK) is pressed.

177. **Height Sorter:** If **I0.0** (High) trigger **Q0.0**; if Low trigger **Q0.1**.

178. **Smart Parking:** Open Gate **Q0.0** only if capacity **C1 < 50**.

179. **Analog Thresholds:** Trigger **Q0.0** if **AIW0 > 10000** and **Q0.1** if **AIW0 > 25000**.

180. **Shift Logger:** Save production count to **VW200** at the end of every hour.

181. **Heater Safety:** Heater **Q0.0** ON only if Overheat sensor **I0.1** is healthy (NC).

182. **Interlocked Motors:** M2 cannot start unless M1 is running. If M1 stops, M2 must stop.

183. **Single Button Start/Stop:** Use one button to toggle the system state.

184. **Pedestrian Crossing:** Modify Traffic Light cycle when button **I0.3** is pressed.

185. **Pressure Control:** Motor ON when Low (**I0.0**), Motor OFF when High (**I0.1**).

186. **Auto Lubrication:** Run Pump **Q0.5** for 10s every 30 minutes of machine run time.

187. **Flow Calculation:** Calculate Flow Rate based on pulse frequency in **VW500**.

188. **Weight Sorting:** Route packages based on weight value in **VW100**.

189. **Door Interlock:** Ensure two doors **Q0.0** and **Q0.1** cannot be open at the same time.

190. **Manual Override:** Allow manual control via **I0.1/I0.2** only when Auto mode is OFF.

191. **Data Logging:** Copy **VW100** to **VW200** upon a trigger pulse.

192. **Fault Threshold:** Sound alarm only after the 3rd consecutive fault occurs.

193. **3-Level Tank:** Control two pumps based on Low, Medium, and High sensors.

194. **Delayed Exhaust:** Fan **Q0.5** runs for 2 mins after Motor **I0.1** stops.

195. **Load Shedding:** Limit system to run only 2 out of 4 motors at any time.

196. **Day/Night Logic:** Security lights ON only when Light Sensor **I0.6** is active.

197. **Baggage Scanner:** Measure bag length using time and reject if too long.

198. **Car Wash Cycle:** Sequential Spray (5s) -> Brush (10s) -> Dry (5s).

199. **Safety Curtain:** Reset system instantly if light curtain **I0.0** is breached.

200. **Final Challenge:** Design a full Industrial Plant Logic including Start/Stop, Emergency, Batch Counting, Fault Logging, and Retentive Memory.